

IME Division

Weekly Portfolio Intelligence

AI-Powered Reporting Automation | dena — Deutsche Energie-Agentur

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The Problem

Every Monday morning, the IME Division Themenbereichsleitung manually consolidates status updates from 25 energy transition projects across 4 data sources — SAP Finance, HR Timesheets, Milestone Tracker, and Project Lead inputs. This takes 1–2 hours per week. Red flags are missed. Leadership receives reports too late to act.



1–2 hours

manual effort every Monday



No early warning

for budget overruns



25 projects, 4 sources,

zero automation

Why AI? Pattern recognition across 4 data sources simultaneously — harder if manually at scale.

The AI Solution

An automated AI agent that reads 25 project data points, validates data quality, generates a leadership report, and delivers it every Monday at 08:00 — without human intervention.



Capability used: NLP generation + structured JSON output + agentic pipeline (Plan → Act → Observe → Iterate)

Python

Claude API

Railway

Resend

NewsAPI

LangSmith

matplotlib

Recording

► **python main.py --now**

Full pipeline: read → validate → report → charts → news → email | ~50–60 seconds

Tools Used

- Python 3.11
- Claude Sonnet 4.6
- Railway (deploy)
- Resend (email)
- NewsAPI
- LangSmith

What the POC Proves

- End-to-end automation works
- AI analysis is accurate
- Email delivery reliable
- Data validation catches errors
- Charts auto-generated

What Production Adds

- Human approval workflow
- Real SAP integration
- Role-based access control
- RAG with BEG/GEG regulations
- Cross-model evaluation

Business Case — ROI

Item	Year 1	Year 3
Upfront investment	€ 2,970	—
Annual operating cost	€ 2,575	€ 2,575
Time saved (hours/year)	142 h*	142 h*
Value created (€45/h)	€ 6,390	€ 6,390
NET ROI	29 %	168 %

Q4 2026

Break-even (conservative)

€45/h

Conservative rate (TVöD E14)

52/year

API calls — minimal carbon footprint

**includes time saving not just by Themenbereichsleitung yet also Finance and Leadership Team*

Top Risks & Mitigations

Risk	Likelihood	Impact	Score	Mitigation
Data quality issues	3	4	12	AI validator catches errors before every report generation
Claude API downtime	2	5	10	Fallback email + exponential retry logic deployed
LLM self-preference bias	3	3	9	Cross-model evaluator (GPT-4o) planned for production
GDPR — US data transfer	2	4	8	EU endpoints active; SCCs verification in progress

✓ These risks are manageable — mitigations are already partially deployed.

Compliance Summary

EU AI Act

LIMITED RISK

Internal management tool.
No decisions about natural persons.
No Annex III context applies.

- Transparency to users
- Human oversight (mandatory)
- Logging & audit trail

GDPR

PRE-PRODUCTION COMPLIANT

Legal basis: Art. 6(1)(f) — Legitimate interest
No special category data processed

 **Open item: SCCs for Resend/Railway pending verification**

Art. 22 — No automated decisions.
All reports reviewed by Themenbereichsleitung before distribution. ✓

Deployment Roadmap

Phase 1 — POC

COMPLETE ✓

- Deployed on Railway
- 25 projects tracked
- Monday 08:00 delivery
- LangSmith eval: 4.78/5

Phase 2 — Pilot

Weeks 1–8

- Real SAP data integration
- 3 Themenleiter as test users
- Human-in-the-loop approval
- DPIA finalized

Phase 3 — Full Deploy

Month 6+

- All IME Division projects
- RAG with BEG/GEG regulations
- Federated to other dena divisions
- News API expanded

Commercialisation

Internal tool → Federated platform across dena Geschäftsbereiche

Success Metrics

Report accuracy $\geq 90\%$ | Time saved $\geq 1\text{h/week}$ | Zero missed escalations

Working MVP — Beyond the POC



AI Data Validator

Claude checks data quality before every report — completeness, plausibility, status consistency



Chart Agent

Risk donut + budget utilization bars auto-generated from live Excel data



News API Integration

6 keyword searches, 7-day window, keyword-based relevance scoring, sorted by score



LangSmith Evaluation

4.78 / 5.00 mean score across 12 projects — completeness, risk clarity, professional tone

Deployed on Railway | Scheduled every Monday 08:00 UTC | Full pipeline: ~50–60 seconds end-to-end

The Recommendation



Business value: 142 hours saved per year, 29% ROI in Year 1, break-even Q4 2026.



Compliance: Limited Risk under EU AI Act. GDPR pre-production compliant. One open item documented.



Next step: Approve a 60-day pilot with real SAP data and 3 Themenleiter as test users.

This system is built, deployed, and ready to pilot.